

Calibration and policy: a clear comparison

Decision readout for the September 14 presentation

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5 September 2026

The result to take into our discussion

The verified candidate improves calibration fit and preserves the policy ordering. The unchanged twelve-target loss falls from **30.483 to 23.153**, a **24.04%** improvement. The largest change in the reported impact or 2063 birth response is **0.090 percentage points**. The candidate remains a review point; the retained production calibration has not been overwritten.

Supply expansion increases housing use and births. Family mortgage credit increases ownership with little change in housing use or births. An unrebated property-tax increase reduces births. Equal tax rebates alter that conclusion because they change household resources. The fiscal convention must accompany every tax claim.

The main calibration weakness remains visible: first-birth rooms are **0.466 versus 0.720** in the data; mean rooms are **6.329 versus 5.780**. The completed empirical check supports retaining the 0.720 target. A better objective value does not resolve these housing discrepancies or establish that the policy fertility responses are directly identified by causal cost variation.

My recommendation is to use this candidate as the working presentation point after our review, with these limitations explicit. Keep the current benchmark as the documented comparison. Another broad calibration search is not the immediate priority.

How much the policy numbers change

Entries below are percent changes in **births per household** against each calibration's own no-policy path. 2063 is a finite transition date.

Policy	Current 2023	Candidate 2023	Current 2063	Candidate 2063
Supply +20%	+1.291	+1.343	+2.050	+2.141
Family LTV 95%	+0.018	+0.018	+0.076	+0.075
Supply and family credit	+1.314	+1.366	+2.155	+2.244
Tax 1% to 2%, no rebate	-1.100	-1.145	-1.623	-1.699

What moves on impact

Candidate policy	Housing user cost	Rooms / HH	Ownership	Births / HH
Supply +20%	-18.192%	+5.743%	+1.836 pp	+1.343%
Family LTV 95%	+0.060%	+0.044%	+0.732 pp	+0.018%
Supply and family credit	-18.017%	+5.888%	+2.988 pp	+1.366%
Tax 1% to 2%, no rebate	+14.577%	-4.923%	-2.334 pp	-1.145%

The supply experiment shifts the supply schedule by 20% at every given price. Equilibrium prices adjust, so the resulting increase in occupied rooms is the smaller amount reported above. Family credit applies to households with dependent children, not only first-time buyers.

For a concrete scale, 10,000 model households have about **846.1 births** in the baseline four-year period. Supply expansion adds about **11.4 births**; family credit adds about **0.15**. This is an illustration of model units, not an annual national forecast.

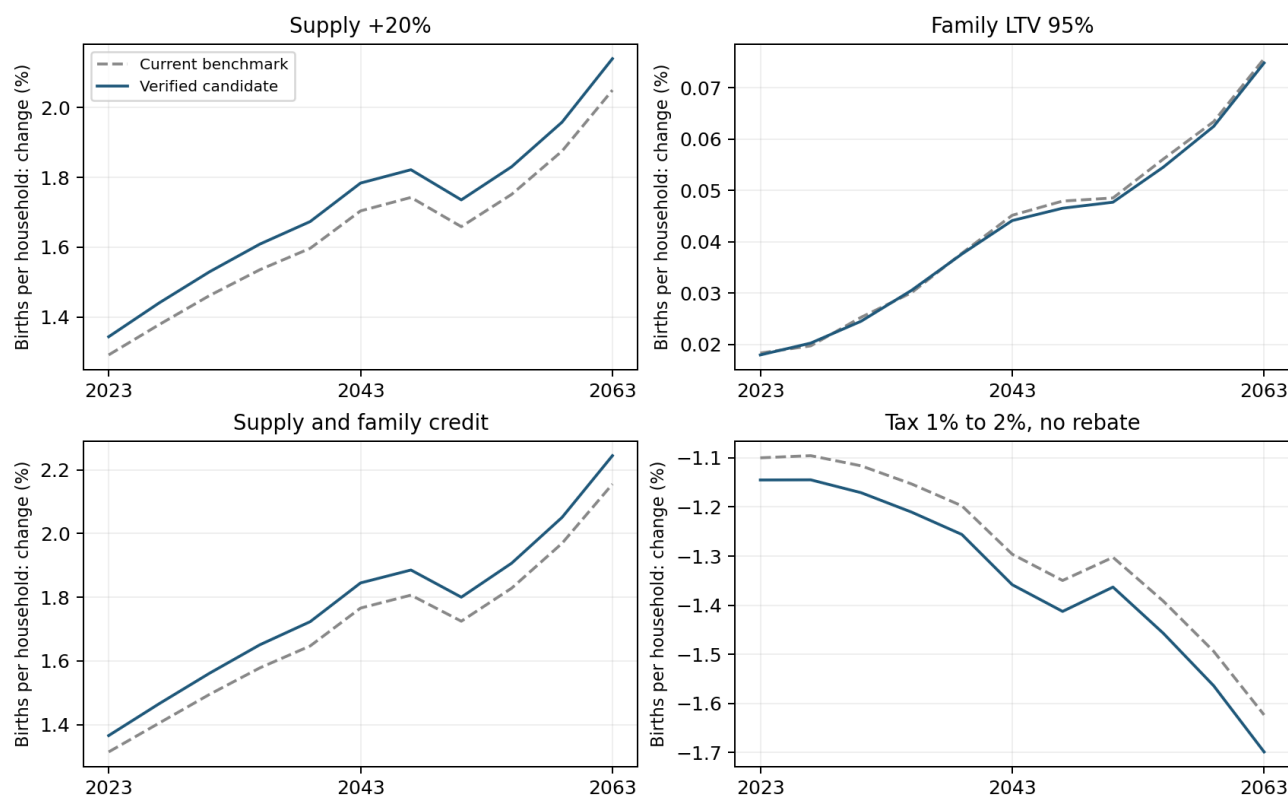


Figure 1: Supplemental comparison of the complete policy paths. Dashed grey is the retained benchmark; solid blue is the candidate. The standard seventeen model diagnostics are preserved separately for every candidate policy date.

Tax results depend on the rebate comparison

The following three-factor Shapley decomposition averages each channel's contribution across all six orders in which tax, house price and rebate can change. Entries are percent of births per household in the **rebated 1%** baseline; the reform is **rebated 2%**. Both equilibria use the same inherited 2023 distribution. The eight component cells reproduce the endpoints and add up.

Birth contribution	Current benchmark	Verified candidate
Direct tax	-1.323%	-1.377%
House-price capitalization	+0.291%	+0.302%
Equal rebate	+1.531%	+1.574%
Net	+0.498%	+0.500%

This net effect is not the effect relative to the unrebated status quo. The five-policy table instead compares unrebated 2% with unrebated 1%. The earlier September 4 chat explicitly drew this distinction; the present review preserves it.

Birth rates and total births in 2063

Candidate policy	Births / HH	Household mass	Total births
Supply +20%	+2.141%	+0.418%	+2.568%
Family LTV 95%	+0.075%	+0.008%	+0.082%
Supply and family credit	+2.244%	+0.428%	+2.682%
Tax 1% to 2%, no rebate	-1.699%	-0.327%	-2.020%

Total births combine the birth rate and the number of households. The exact identity is checked for every row. Household mass is identical across policies before 2043, as required by the maintained twenty-year entry lag. The inherited household queue is a conditional mechanism closure; this table is not a validated resident-population forecast or a terminal steady-state comparison.

What was checked, and what remains open

All five short transition tests passed before the full five paths were launched. Every full path exactly reproduces its short test's first two dates. The historical reconstruction, shared starting state, market clearing, population, feasibility, probability and accounting checks pass at the existing tolerances. Each of the 55 policy dates has a saved state and the unchanged seventeen-graph packet.

The source contains the already verified policy-reuse correction. Its fixed- parameter history reproduces all twelve target rows, every parameter and bound, and 253 numeric historical entries of the original candidate exactly. No model primitive, target, weight, parameter or numerical tolerance was changed here.

The numerical screen asks whether household value ever falls when wealth rises. It flags **28 of 55 dates**, but the largest affected share is only **1.77 per million** of the inherited household distribution. For three flagged policy-date distributions, a six-solve check first reproduces the saved policies exactly, then searches the full saving grid at the same prices. This removes the flagged value decreases in all three tests. The largest absolute change in births per household is **0.000346%**. The eleven tax equilibria/ component cells contain 4 flagged packets, with maximum affected share **2.58 per million**. These checks do not bound the error of a fully re-solved equilibrium path or the tax decomposition. The largest share affected by the reporting consumption floor exceeding the budget is **0.022 per million**. Both rebated equilibria also pass the check that the selected rebate matches the parameter state used for accounting.

The old chats explain several apparent inconsistencies. Supply elasticity 1.75 initializes the old equilibrium; 0.63 applies to dated markets. This distinction was already recorded, though its economic justification remains open. A separate, later demographic branch tracks people more consistently; it does not change this retained household-entry experiment. The rebated-tax result uses its own fiscal baseline. Direct author statements support studying policy during the transition, without settling every inherited population assumption.

For the presentation, the defensible claim is that these mechanisms survive a meaningful improvement in fit under the same target system. The remaining work is to decide how prominently to qualify the housing-fit shortfall, population closure and fiscal scope. This readout does not certify a new perfect-foresight equilibrium, welfare ranking or globally optimal calibration.

Complete target fit: verified candidate

Gaps equal model minus target. Contributions are weight times squared gap. The weights retain their existing empirical or synthetic interpretations; this loss is not a specification-test statistic.

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918000	1.921698	+0.003698	1425.74	0.0195
Childlessness	0.188000	0.189859	+0.001859	17180.7	0.0594
Mean age at first birth	26.044627	26.251703	+0.207076	44.4444	1.9058
First births at age 30+	0.260327	0.237163	-0.023165	10000	5.3660
Rooms response to first birth	0.720246	0.466044	-0.254202	137.565	8.8893
Rooms, 3+ minus 1-2 children	0.367700	0.379198	+0.011499	2958.51	0.3912
Family ownership gap	0.167662	0.169578	+0.001917	14229.6	0.0523
Ownership, ages 30-55	0.575472	0.546618	-0.028854	1207.85	1.0056
Mean occupied rooms	5.779970	6.328594	+0.548624	11.9732	3.6038
Wealth / annual earnings	6.873100	7.031196	+0.158096	6.28767	0.1572
Bequest flow / wealth	0.008800	0.008485	-0.000315	5.16529e+06	0.5139
Older wealth/income p90 / p50	3.448111	3.303595	-0.144516	56.9598	1.1896

Complete target fit: retained production

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918000	1.922907	+0.004907	1425.74	0.0343
Childlessness	0.188000	0.189407	+0.001407	17180.7	0.0340
Mean age at first birth	26.044627	26.256032	+0.211404	44.4444	1.9863
First births at age 30+	0.260327	0.237579	-0.022748	10000	5.1748
Rooms response to first birth	0.720246	0.436418	-0.283828	137.565	11.0821
Rooms, 3+ minus 1-2 children	0.367700	0.403441	+0.035741	2958.51	3.7793
Family ownership gap	0.167662	0.161699	-0.005963	14229.6	0.5060
Ownership, ages 30-55	0.575472	0.544488	-0.030984	1207.85	1.1596
Mean occupied rooms	5.779970	6.317291	+0.537321	11.9732	3.4568
Wealth / annual earnings	6.873100	6.932652	+0.059552	6.28767	0.0223
Bequest flow / wealth	0.008800	0.008433	-0.000367	5.16529e+06	0.6971
Older wealth/income p90 / p50	3.448111	3.236511	-0.211600	56.9598	2.5503

Complete parameter comparison

Parameter	Production	New candidate	Bounds / restriction	Near bound?
Annual patience β	0.995117	0.995739	[0.94, 0.9995]	No
First-birth dispersion κ_1	2.168173	2.168173	[0.02, 50]	No
Later-birth dispersion κ_C	1.736471	1.736471	[0.02, 50]	No
Owner-service premium χ	1.043472	1.043472	[0.1, 5]	No
Supply intercept H_0	14.562959	14.562959	[0.2, 80]	No
Bequest strength θ_0	0.528428	0.549189	[0, 8]	No
Bequest shifter θ_1	0.107249	0.107249	[0.02, 16]	Raw scale only
Per-child room floor	0.282210	0.248877	[0.1, 1.8]	No
First-birth fixed cost	4.559138	4.559138	[0, 8]	No
First-child room intercept	0.364931	0.464931	[0, 0.5]	No
2007-2023 child-value change	-0.328714	-0.328714	[-1.5, 0.2]	No
Tenure dispersion κ	0.005000	0.005000	Externally fixed	Not estimated
Dated supply elasticity η	0.630000	0.630000	Externally fixed	Not estimated

The old fertility-preference intercept is re-normalized to replacement for every parameter vector. New candidate values are $\psi_{2007} = 0.28947879$ and $\psi_{2023} = -0.03923512$. The stored near-bound flag uses raw parameter units. For the bequest shifter, interpret it alongside its position in the logarithmic search interval; a raw-scale flag alone does not establish a binding search constraint.

The supply restriction 0.63 applies to dated markets. The inherited initialization retains elasticity 1.75 before the dated supply intercept is normalized. Both values and that order are unchanged across this comparison; whether this is the intended economic contract remains a discussion item in the separate code audit.

Reproduction and evidence

The evidence folder is `output/model/e5f_candidate_policy_comparison_20260905a`. It contains the pinned plan, historical reconciliation, source and input hashes, all policy rows, exact smoke/full comparison receipts, tax decomposition, numerical-flag check and complete fit/parameter tables. Its README states the run budgets and commands. The new comparison CSV contains all 110 policy/year/calibration rows, including the no-policy paths. The standard graphs are in `full_diagnostics/`, grouped by policy and date. All paths are relative to: `/Users/tommasodesanto/Desktop/Projects/Fertility/Fertility_Spring26`.

Rebuild this readout from the collected outputs:

```
python3 code/model/tools/build_e5f_candidate_policy_comparison_report.py
```